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Robustness of Brain Tumor Segmentation under MRI Corruptions

Project Report

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Abstract

This report investigates the robustness of a pretrained MONAI SegResNet model for BraTS 2020 brain tumor segmentation under FLAIR modality corruptions. Two corruption families were evaluated: additive Gaussian noise and multiplicative bias field distortion. Three mitigation strategies were compared: linear preprocessing filters, N4 bias-field correction, and corruption-aware fine-tuning. Experiments used a fixed train-validation-test split of 258/55/55 patients and paired evaluation on the same 55 test patients. Gaussian noise reduced mean WT-DSC from 0.914 to 0.829, but a simple `mean_3` filter recovered performance to 0.905. Bias field corruption reduced mean WT-DSC more strongly to 0.766, and linear filters failed to recover performance. N4 correction and bias-augmented fine-tuning both achieved approximately 0.820 mean WT-DSC and were statistically tied. The results show that robustness mitigation must match the physical structure of the artifact: smoothing is effective for high-frequency additive noise, whereas structured multiplicative intensity distortion requires artifact-specific correction or model adaptation.

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Chapter 1

Introduction

Automatic brain tumor segmentation is an important task in computer-aided medical image analysis. In clinical and research workflows, segmentation models are often evaluated on clean benchmark data, although real magnetic resonance imaging (MRI) acquisitions may contain acquisition artifacts. Such artifacts can alter the intensity distribution of one modality and may reduce the reliability of a segmentation model, especially when the model was trained or pretrained under cleaner conditions.

This project investigates the robustness of a pretrained brain tumor segmentation model under controlled corruptions of the FLAIR MRI modality. The central research question is:

> What happens to brain tumor segmentation performance when the FLAIR modality is corrupted, and which mitigation strategy is most effective for different corruption types?

Two representative corruption families were studied. First, additive Gaussian noise was used to model high-frequency random intensity disturbances, for example from scanner noise, weak signal, fast acquisition, or patient motion. Second, a multiplicative bias field was used to model a smooth low-frequency intensity inhomogeneity, such as those caused by magnetic field inhomogeneity, coil sensitivity differences, or acquisition setup. These two corruptions were selected because they represent qualitatively different failure modes: Gaussian noise is local and high-frequency, whereas bias field corruption is spatially smooth, structured, and multiplicative.

The project compares three mitigation strategies. The first strategy applies simple linear preprocessing filters to the corrupted FLAIR image before inference. The second strategy applies N4 bias-field correction, a specialized preprocessing method designed for multiplicative MRI intensity inhomogeneity. The third strategy uses model-level adaptation through fine-tuning with corruption-specific FLAIR augmentation. All methods are evaluated on the same held-out test patients to allow a paired comparison across conditions.

Chapter 2

Background

2.1 Brain Tumor Segmentation in Multimodal MRI

Glioma segmentation is commonly performed using multiple MRI modalities because each sequence emphasizes different anatomical and pathological properties. In this project, each patient volume contains four co-registered MRI modalities: T1, contrast-enhanced T1 (T1ce), T2, and FLAIR. The model receives all four modalities as input and predicts tumor masks for three nested tumor regions: whole tumor (WT), tumor core (TC), and enhancing tumor (ET).

FLAIR is particularly important for detecting edema and the broader tumor extent. Therefore, corruption of the FLAIR modality can directly affect WT segmentation and may also influence the derived TC and ET predictions through the multi-channel model input. For this reason, the robustness experiments focus on corrupting only the FLAIR channel while keeping the other input modalities unchanged.

2.2 MRI Corruptions

2.2.1 Gaussian Noise

Gaussian noise was implemented as an additive corruption of the FLAIR image:

$$I_{\text{corr}}(x) = I_{\text{FLAIR}}(x) + \epsilon(x), \qquad \epsilon(x) \sim \mathcal{N}\Big(0,\ \sigma_{\text{rel}}^2 \cdot \text{std}(I_{\text{FLAIR,brain}})^2\Big)\,.$$

The noise standard deviation is scaled by the per-patient FLAIR brain standard deviation, so that the same σ_{rel} produces a comparable corruption severity across patients with different absolute intensity scales. A severity sweep was performed over $\sigma_{\text{rel}} \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and the final test experiments used $\sigma_{\text{rel}} = 3.0$. Because the artifact is high-frequency and additive, it is expected that smoothing filters can remove a substantial part of the corruption while preserving the underlying structure.

2.2.2 Multiplicative Bias Field

The bias field corruption was implemented as a multiplicative spatial intensity distortion:

$$I_{\text{corr}}(x) = I_{\text{FLAIR}}(x) \cdot b(x),$$

where $b(x)$ is a smooth, spatially varying field. The field was generated as a sum of randomly placed three-dimensional Gaussian blobs, normalized and clipped according to the selected strength. The background was kept at zero, and the same deterministic per-patient seed was used across conditions.

A severity sweep was performed over strengths $\{0.2, \ldots, 1.2\}$, and the final experiments used strength 1.2. Unlike Gaussian noise, the bias field does not simply add random local noise. It changes the relative intensity distribution across the brain. Therefore, simple smoothing is not expected to reverse the distortion and may even blur useful anatomical structures.

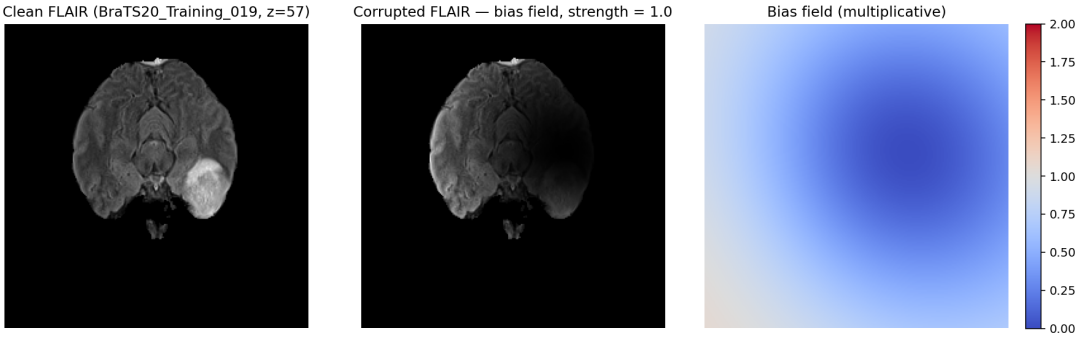


Figure 2.1: Visual example of synthetic multiplicative bias-field corruption. The clean FLAIR image is multiplied by a smooth low-frequency field, producing a spatially non-uniform intensity distortion.

2.3 Mitigation Strategies

2.3.1 Linear Preprocessing Filters

The first mitigation family applies simple filters to the corrupted FLAIR modality before the model receives the input. The evaluated filter bank consisted of no filtering, a 3D mean filter with a 3^3 uniform kernel, and Gaussian filters with standard deviations of 0.5, 1.0, and 1.5 voxels. These filters mainly reduce local high-frequency variations and therefore are expected to be suitable for Gaussian noise but insufficient for structured bias fields.

2.3.2 N4 Bias-Field Correction

N4 correction was included as a specialized mitigation for the bias field experiment. It estimates a smooth multiplicative intensity field and corrects the image by dividing out this estimated field. In the implementation, N4 was applied only to the corrupted FLAIR modality using a brain mask. The correction used a three-level schedule with 20 iterations per level. N4 was not evaluated for Gaussian noise because it is designed for smooth multiplicative intensity inhomogeneity rather than additive noise.

2.3.3 Fine-Tuning with Corruption Augmentation

The model-level mitigation strategy fine-tunes the pretrained segmentation model using corruption-specific augmentation. Fine-tuning started from the pretrained MONAI bundle weights and used AdamW with a learning rate of 10^{-5}. The loss function was DiceCELoss with sigmoid activation. FLAIR-only corruption augmentation was applied with probability 0.7. For Gaussian

fine-tuning, σ was sampled from $U(0.2, 3.0)$. For bias-field fine-tuning, the strength was sampled from $U(0.2, 1.2)$. Fine-tuning was run for 3000 steps, with validation used for checkpoint selection.

Chapter 3

Materials and Methods

3.1 Dataset

The experiments used the BraTS 2020 training data obtained from the Kaggle BraTS2020 dataset page [1]. BraTS is a standardized benchmark for multimodal brain tumor segmentation and is described by Menze et al. [2]. The dataset contained 368 patients with full ground-truth segmentation. For every patient, four co-registered modalities were available: T1, T1ce, T2, and FLAIR. The segmentation labels were evaluated as three nested regions: WT, TC, and ET.

The data were split into training, validation, and test sets. The training set contained 258 patients, corresponding to 70% of the dataset. The validation set contained 55 patients, corresponding to 15%, and was used for severity selection and checkpoint selection. The test set also contained 55 patients and was used only for final evaluation. The same 55 test patients were used across all conditions, and deterministic per-patient corruption seeds were used to ensure that paired comparisons were meaningful.

3.2 Segmentation Model

The experiments used the pretrained MONAI `brats_mri_segmentation` bundle. MONAI is an open-source PyTorch-based framework for medical imaging AI, and the official project website was used as the main software reference [3]. The MONAI framework is also described in the paper by Cardoso et al. [4]. The model architecture was SegResNet with pretrained MONAI weights. The input consisted of the four MRI modalities T1, T1ce, T2, and FLAIR. The output consisted of predicted tumor masks for WT, TC, and ET.

Before applying corruptions, the pretrained model was evaluated on clean test inputs. The clean baseline on the 55-patient test set achieved a WT Dice score of 0.914, a TC Dice score of 0.886, and an ET Dice score of 0.766. This clean baseline served as the reference point for measuring robustness degradation and mitigation recovery.

3.3 Experimental Design

The evaluation followed a paired design. For each test patient, the model was evaluated under clean input, corrupted input, and multiple mitigation conditions. Because the same patients were used in each condition, differences between conditions could be analyzed using paired statistics.

For Gaussian noise, the final severity was $\sigma = 3.0$. The evaluated conditions included the clean

pretrained model, corrupted input without filtering, linear filters, corruption-specific fine-tuning, and clean input evaluated with the fine-tuned model as a forgetting check.

For bias field corruption, the final severity was strength 1.2. The evaluated conditions included the clean pretrained model, corrupted input without filtering, linear filters, N4 correction, bias-augmented fine-tuning, and clean input evaluated with the bias-fine-tuned model as a forgetting check.

3.4 Evaluation Metrics

The primary overlap metric was the Dice similarity coefficient (DSC):

$$\mathrm{DSC} = \frac{2|P \cap G|}{|P| + |G|},$$

where P is the predicted tumor region and G is the corresponding ground-truth region. DSC ranges from 0 to 1, where higher values indicate better overlap.

The second main metric was the 95th percentile Hausdorff distance (HD95), measured in millimeters. HD95 evaluates boundary error while being more robust to outliers than the maximum Hausdorff distance. Lower HD95 values indicate better boundary agreement. Additional metrics were also computed to provide a more complete evaluation. Sensitivity measures the fraction of ground-truth tumor voxels that were detected. False-positive volume (FPvol) measures the amount of predicted tumor outside the ground truth in milliliters, and false-negative volume (FNvol) measures the amount of missed tumor volume in milliliters. Therefore, DSC describes overlap, HD95 describes boundary quality, and FPvol/FNvol describe whether the model tends to over-segment or miss tumor tissue.

For statistical comparison, paired two-sided Wilcoxon signed-rank tests were applied to WT-DSC values over the 55 test patients. WT-DSC was used for the headline comparisons because FLAIR corruption most directly affects the whole tumor region.

Chapter 4

Results

4.1 Severity Selection

The validation sweep confirmed that the two corruption families behave differently. For Gaussian noise, raw corrupted performance decreased as σ increased, but smoothing filters remained close to the clean baseline. Therefore, the highest tested severity, $\sigma = 3.0$, was selected for the final experiment.

For bias field corruption, the raw corrupted condition also degraded as severity increased. However, the linear smoothing filters did not recover the distortion and often tracked the corrupted condition closely. N4 correction remained the most stable preprocessing method during the sweep. The final selected bias strength was 1.2, where the corruption was clearly harmful and N4 still left enough residual gap for fine-tuning to compete.

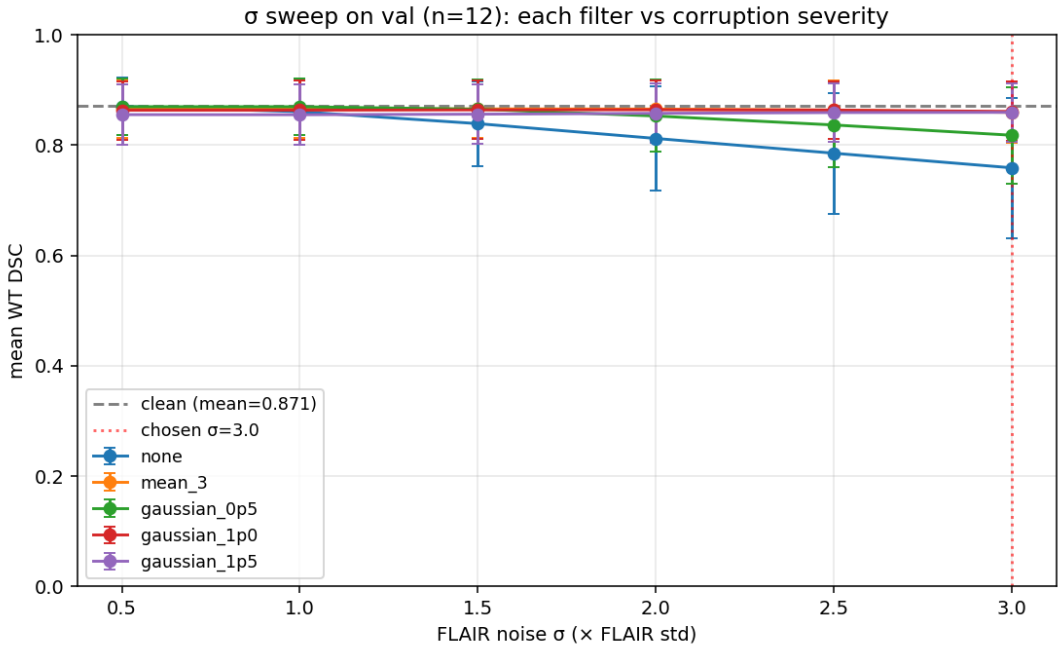


Figure 4.1: Validation severity sweep for Gaussian noise. The final severity was selected as $\sigma = 3.0$, where the corruption is strong but smoothing filters can still recover most of the WT-DSC.

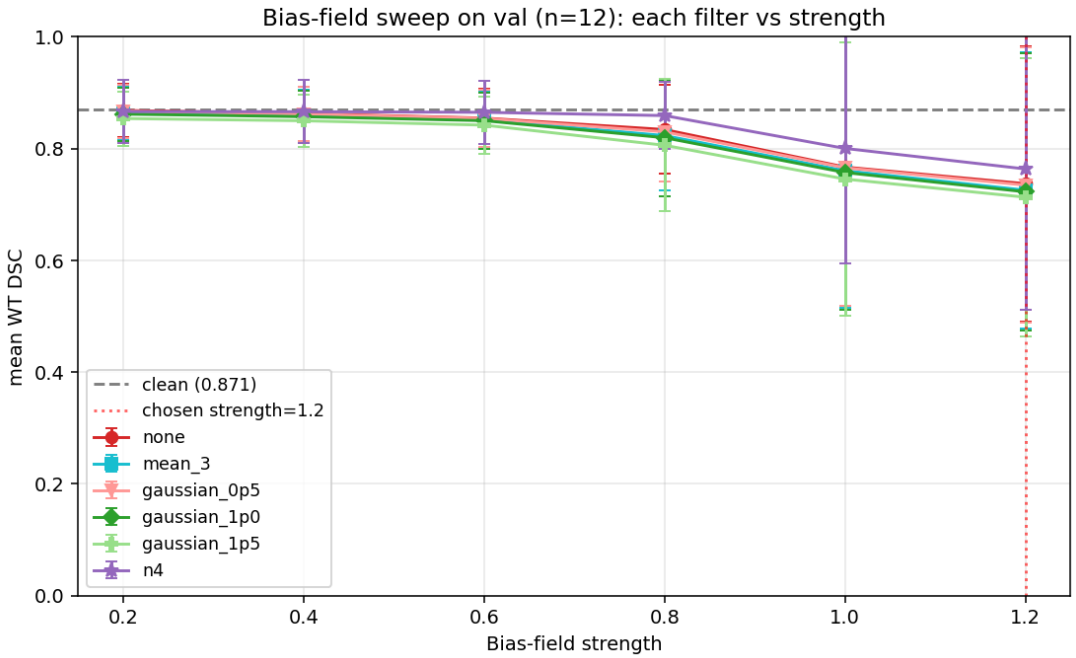


Figure 4.2: Validation severity sweep for the multiplicative bias field. The final severity was selected as strength = 1.2, where N4 remained the most stable correction.

4.2 Clean Baseline

On the clean 55-patient test set, the pretrained model achieved strong baseline performance. The mean WT-DSC was 0.914, the mean TC-DSC was 0.886, and the mean ET-DSC was 0.766. These values represent the reference performance before introducing FLAIR corruption.

4.3 Gaussian Noise Results

Gaussian noise reduced the mean WT-DSC from 0.914 to 0.829, corresponding to a drop of 8.5 percentage points. Linear filtering recovered most of this degradation. The best linear filter, `mean_3`, achieved a mean WT-DSC of 0.905, only 0.9 points below the clean baseline. The Gaussian filter with $\sigma = 1.0$ achieved a similar mean WT-DSC of 0.905.

Fine-tuning also improved performance compared with the raw corrupted condition, reaching a mean WT-DSC of 0.878. However, it remained below the best linear filter. This indicates that for additive high-frequency Gaussian noise, direct preprocessing is more effective than forcing the model to internally adapt to noisy inputs.

Table 4.1: Headline WT-DSC results for Gaussian noise at $\sigma = 3.0$.

Condition	Mean WT-DSC	Interpretation
Clean pretrained	0.914	Clean reference
Corrupted pretrained	0.829	Robustness drop
Best linear filter: `mean_3`	0.905	Near-full recovery
Fine-tuned on corrupted input	0.878	Partial recovery
Clean fine-tuned	0.894	Forgetting check

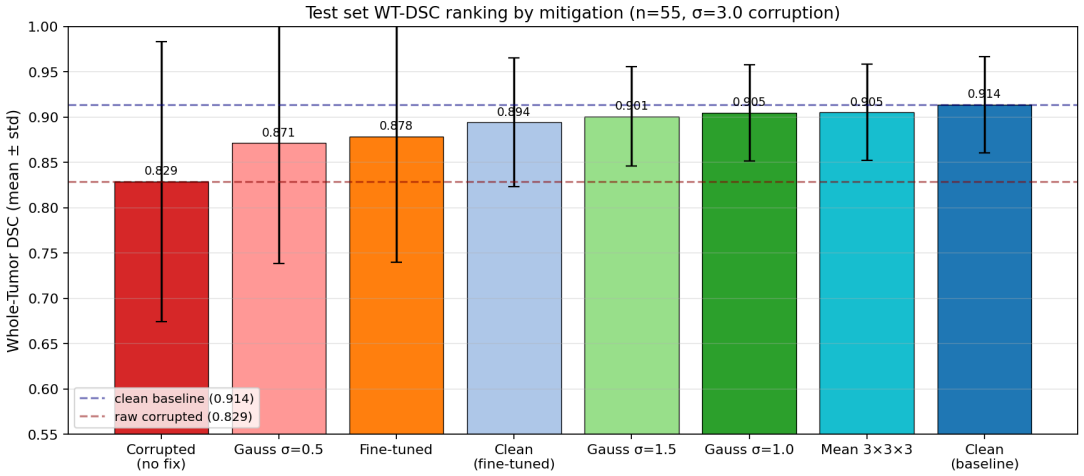


Figure 4.3: Test-set WT-DSC ranking under Gaussian noise corruption. Linear filtering recovers most of the lost performance, while fine-tuning is slightly behind the best preprocessing methods.

The Wilcoxon comparison showed that the corrupted condition was significantly worse than the clean baseline, with a median WT-DSC difference of -0.0646 and $p = 3.70 \times 10^{-8}$. The best filter significantly improved over the corrupted condition, with a median difference of $+0.0527$ and $p = 1.05 \times 10^{-8}$. Fine-tuning was statistically worse than the best filter, with a median difference of -0.0050 and $p = 2.35 \times 10^{-3}$. Clean performance after Gaussian fine-tuning dropped by 2.0 points in mean WT-DSC (paired Wilcoxon median difference -0.0124, $p = 2.91 \times 10^{-8}$), indicating statistically significant clean-domain forgetting.

4.4 Bias Field Results

Bias field corruption produced a stronger degradation than Gaussian noise. Mean WT-DSC dropped from 0.914 to 0.766, corresponding to a drop of 14.8 percentage points. The linear filters failed to recover the bias distortion. The best linear filter on the test set was `gaussian_0p5`, but it reached only 0.760 mean WT-DSC, slightly below the raw corrupted condition. Wider smoothing filters performed even worse.

N4 correction and bias-augmented fine-tuning achieved the best recovery. Both methods reached approximately the same mean WT-DSC of 0.820–0.821. N4 achieved a median WT-DSC of 0.889, while fine-tuning achieved a median WT-DSC of 0.861. The paired Wilcoxon comparison between fine-tuning and N4 showed no significant difference, with a median difference of -0.0005 and $p = 0.987$. Therefore, the two strategies were statistically tied under the selected bias severity.

Table 4.2: Headline WT-DSC results for bias field corruption at strength 1.2.

Condition	Mean WT-DSC	Interpretation
Clean pretrained	0.914	Clean reference
Corrupted pretrained	0.766	Strong robustness drop
Best linear filter: `gaussian_0p5`	0.760	No recovery
N4 correction	0.820	Best data-level correction
Fine-tuned on corrupted input	0.820	Best model-level correction
Clean fine-tuned	0.909	Forgetting check

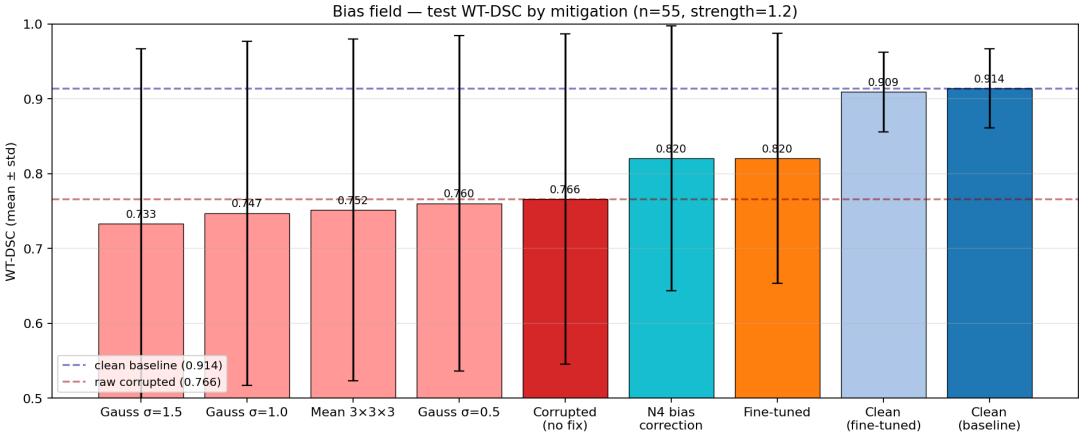


Figure 4.4: Test-set WT-DSC ranking under bias-field corruption. Linear filters fail to recover the structured distortion, while N4 correction and fine-tuning perform best.

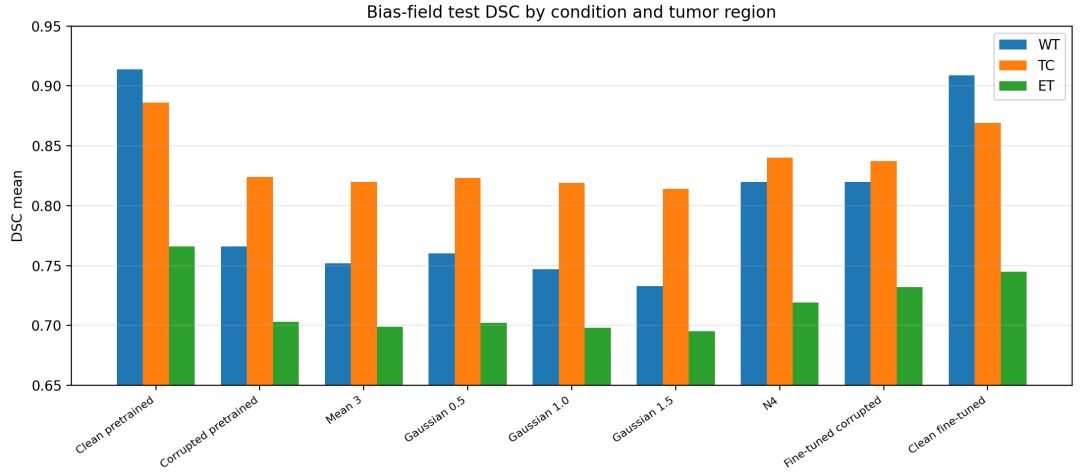


Figure 4.5: DSC by tumor region under bias-field corruption. WT, TC, and ET are affected differently, showing that robustness should not be judged only from a single tumor region.

The statistical results support this interpretation. Bias corruption was significantly worse than clean input, with a median WT-DSC difference of -0.0478 and $p = 4.72 \times 10^{-9}$. The best linear filter was significantly below the raw corrupted input, with a median difference of -0.0021 and $p = 1.94 \times 10^{-7}$, confirming that simple smoothing is not a useful mitigation for bias fields. N4 significantly improved over the corrupted condition, with a median difference of $+0.0162$ and $p = 7.94 \times 10^{-8}$. Fine-tuning significantly outperformed the best linear filter, with a median difference of $+0.0232$ and $p = 3.53 \times 10^{-8}$. Clean performance after bias fine-tuning dropped only slightly, from 0.914 to 0.909 mean WT-DSC, corresponding to a forgetting cost of 0.5 points (paired Wilcoxon median difference -0.0038, $p = 5.99 \times 10^{-3}$). The drop is small in magnitude but statistically significant, and it is four times smaller than the forgetting cost observed for the Gaussian-augmented fine-tune.

4.5 Qualitative Analysis

A qualitative example was inspected for a bias-corrupted test patient. This visual comparison helps explain the numerical results. The bias field changes the FLAIR intensity distribution

and can therefore alter the predicted tumor boundaries. Linear smoothing is not sufficient for this type of smooth multiplicative distortion, whereas N4 correction and fine-tuning are better aligned with the artifact type.

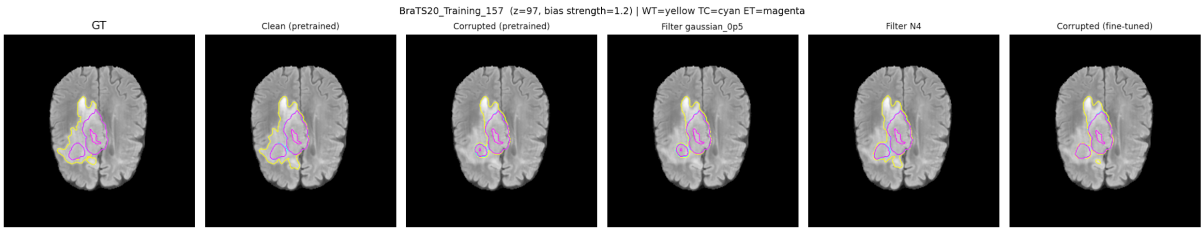


Figure 4.6: Qualitative segmentation example for a bias-corrupted test patient. The figure compares ground truth, clean pretrained prediction, corrupted pretrained prediction, Gaussian-filtered prediction, N4 correction, and corrupted fine-tuned prediction. WT is shown in yellow, TC in cyan, and ET in magenta.

4.6 Additional Evaluation Metrics

Besides DSC, the notebook computed boundary and volume-based metrics. These metrics provide a more detailed view of segmentation failure. HD95 measures whether the predicted tumor boundary is spatially close to the ground truth. FPvol measures how much extra tumor volume was predicted outside the ground truth, and FNvol measures how much tumor volume was missed.

Table 4.3: Additional WT metrics under bias-field corruption. HD95 is measured in millimeters; FPvol and FNvol are measured in milliliters.

Condition	WT HD95 mean	WT FPvol mean	WT FNvol mean
Clean pretrained	5.204	9.972	4.418
Corrupted pretrained	20.671	20.336	18.179
Best linear filter: `gaussian_0p5`	21.143	21.613	18.112
N4 correction	14.006	13.278	15.195
Fine-tuned on corrupted input	16.393	14.083	14.043
Clean fine-tuned	8.333	9.361	5.765

The additional metrics support the DSC-based interpretation. Bias corruption increased WT HD95 from 5.204 mm to 20.671 mm, meaning that the tumor boundary became much less accurate. The best linear filter did not improve this boundary error and remained slightly worse than the raw corrupted input. N4 reduced WT HD95 to 14.006 mm, while fine-tuning reduced it to 16.393 mm. For volume errors, both N4 and fine-tuning reduced false-positive and false-negative volume compared with the corrupted input. Fine-tuning achieved the lowest FNvol among corrupted-input mitigations, suggesting that it missed less tumor volume than N4, while N4 achieved the lowest FPvol and HD95.

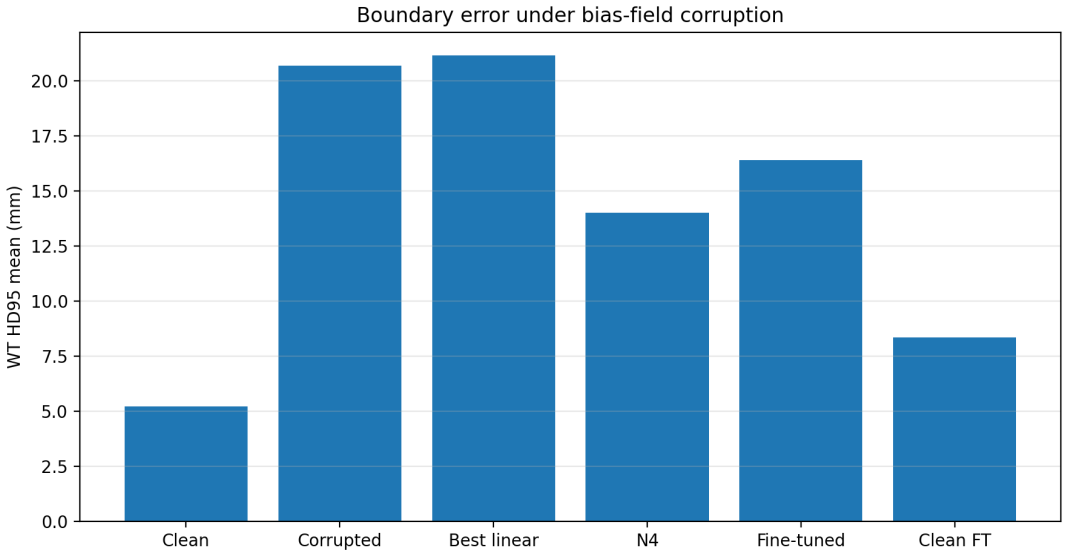


Figure 4.7: WT HD95 under bias-field corruption. Boundary error increases strongly after corruption. N4 gives the best boundary recovery among corrupted-input mitigations.

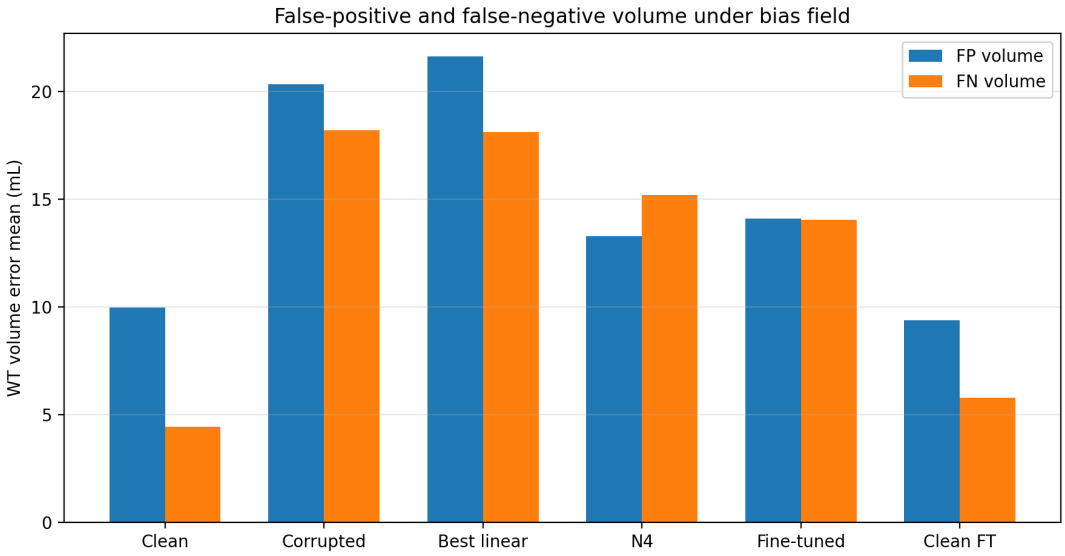


Figure 4.8: WT false-positive and false-negative volume under bias-field corruption. N4 reduces false-positive volume most strongly, while fine-tuning has the lowest false-negative volume among corrupted-input mitigations.

4.7 Comparison Between Corruption Families

The central result of the project is that the best mitigation depends strongly on the type of corruption. Gaussian noise mainly creates high-frequency local artifacts. Therefore, linear smoothing filters can remove much of the corruption while preserving the segmentation-relevant signal. In this setting, the best filter almost fully recovered the clean baseline and outperformed fine-tuning.

Bias field corruption is different. It introduces a smooth global intensity distortion rather than

random noise. Linear smoothing cannot undo the multiplicative field and instead may blur useful structures. Consequently, the linear filters performed at or below the raw corrupted condition. N4 correction and bias-augmented fine-tuning performed best because they address the correct failure mode: N4 explicitly estimates and removes the multiplicative field, while fine-tuning teaches the model to become invariant to such intensity inhomogeneity.

Table 4.4: Side-by-side comparison of headline WT-DSC results.

Condition	Gaussian noise	Bias field
Clean pretrained	0.914	0.914
Corrupted pretrained	0.829	0.766
Best linear filter	0.905	0.760
N4 correction	n/a	0.820
Fine-tuned on corrupted input	0.878	0.820
Clean fine-tuned	0.894	0.909

This comparison shows that a universal mitigation strategy is not sufficient. The same linear filter bank that nearly solved Gaussian noise failed against bias field corruption. Conversely, specialized correction and augmentation were necessary for the structured corruption. The results therefore highlight the importance of matching the mitigation strategy to the physical nature of the artifact.

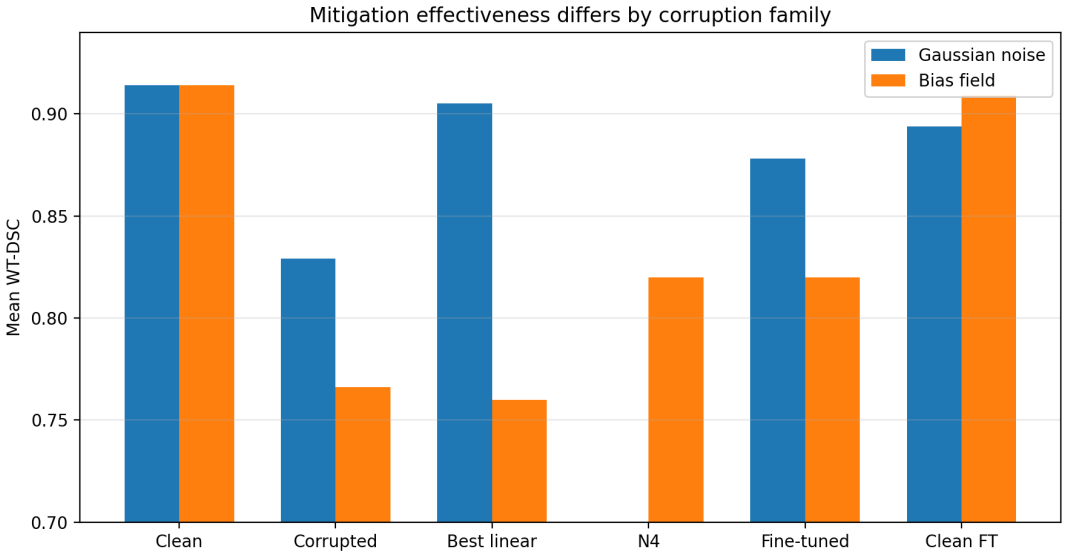


Figure 4.9: WT-DSC comparison between Gaussian noise and bias-field corruption. Linear filtering recovers Gaussian noise well, while N4 correction and fine-tuning are needed for bias-field corruption.

Chapter 5

Discussion

5.1 Main Findings

The experiments produced three main findings. First, FLAIR corruption can substantially degrade brain tumor segmentation, even when the remaining modalities are clean. The effect was particularly strong for the bias field corruption, where the mean WT-DSC dropped by 14.8 points.

Second, simple linear filtering is effective only for additive high-frequency noise. Under Gaussian noise, the `mean_3` filter recovered the WT-DSC from 0.829 to 0.905, close to the clean baseline of 0.914. Under bias field corruption, however, the best linear filter reached only 0.760 and was worse than the raw corrupted condition. This confirms that smoothing is not a general robustness solution.

Third, N4 correction and fine-tuning were both effective for the bias field. Their performance was statistically tied on WT-DSC. The additional metrics add nuance: N4 produced better boundary recovery and lower false-positive volume, whereas fine-tuning produced slightly lower false-negative volume. Thus, the choice between N4 and fine-tuning may depend on whether boundary precision, over-segmentation, or missed tumor volume is considered most important.

5.2 Forgetting on Clean Inputs

Fine-tuning can improve robustness but may also reduce performance on clean inputs. The forgetting check evaluated the fine-tuned models on clean test data. Gaussian-augmented fine-tuning reduced clean WT-DSC from 0.914 to 0.894, a drop of 2.0 points. Bias-augmented fine-tuning reduced clean WT-DSC only from 0.914 to 0.909, a drop of 0.5 points.

This difference suggests that bias-field augmentation was less harmful to the clean representation learned by the pretrained model. A possible explanation is that smooth multiplicative intensity variation resembles natural MRI acquisition variability more closely than strong additive Gaussian noise. Therefore, bias augmentation may expand the training distribution without forcing the model as far away from clean inputs.

5.3 Clinical and Technical Implications

The results emphasize that robustness evaluation should consider artifact type, not only artifact severity. A method that works well for one corruption can fail for another. For practical medical

image analysis pipelines, this means that preprocessing and adaptation strategies should be chosen based on the expected acquisition artifact. For random high-frequency noise, simple denoising may be sufficient. For intensity inhomogeneity, N4 correction or corruption-aware training is more appropriate.

The results also show the value of paired evaluation. By applying all conditions to the same 55 test patients, performance differences can be attributed more directly to the corruption and mitigation rather than to differences in patient composition. This design makes the comparison between preprocessing and fine-tuning more reliable.

5.4 Limitations

This project has several limitations. First, only two corruption types were evaluated. Real MRI data can contain many other artifacts, including motion, ghosting, ringing, resolution loss, and missing modalities. Second, all experiments used a single pretrained architecture, MONAI SegResNet. The conclusions may differ for other segmentation architectures or training regimes. Third, the bias field corruption was synthetic and closely matched the assumptions of N4 correction. Real scanner artifacts may be more complex. Fourth, the experiments focused mainly on WT-DSC for headline comparisons. Additional HD95 and volume-based results were included, but a full clinical evaluation would require broader external validation and artifact types beyond the synthetic corruptions used here.

Chapter 6

Conclusion

This project evaluated the robustness of a pretrained MONAI SegResNet brain tumor segmentation model under two FLAIR corruption families: additive Gaussian noise and multiplicative bias field corruption. The experiments used the BraTS 2020 dataset with a fixed 258/55/55 train-validation-test split and a paired evaluation on 55 test patients.

The results show that the optimal mitigation strategy depends on the corruption type. For Gaussian noise, simple linear filtering nearly restored clean performance and outperformed fine-tuning. For bias field corruption, linear filters failed, while N4 correction and bias-augmented fine-tuning achieved the best recovery and were statistically tied. Bias-augmented fine-tuning also caused only minimal forgetting on clean inputs.

Overall, the project demonstrates that robustness cannot be judged by a single corruption or a single mitigation method. Effective robustness requires understanding the physical structure of the artifact and selecting a mitigation strategy that directly addresses that structure.

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